VERIDI LAST-MILE DELIVERY AUDIT

Are we failing specific regions, or is this nationwide?

Analysis of 99,441 Olist orders

Delivery estimates are padded, not optimistic

93.2%

of 96,470 delivered orders arrived on or before the promised date

12 days

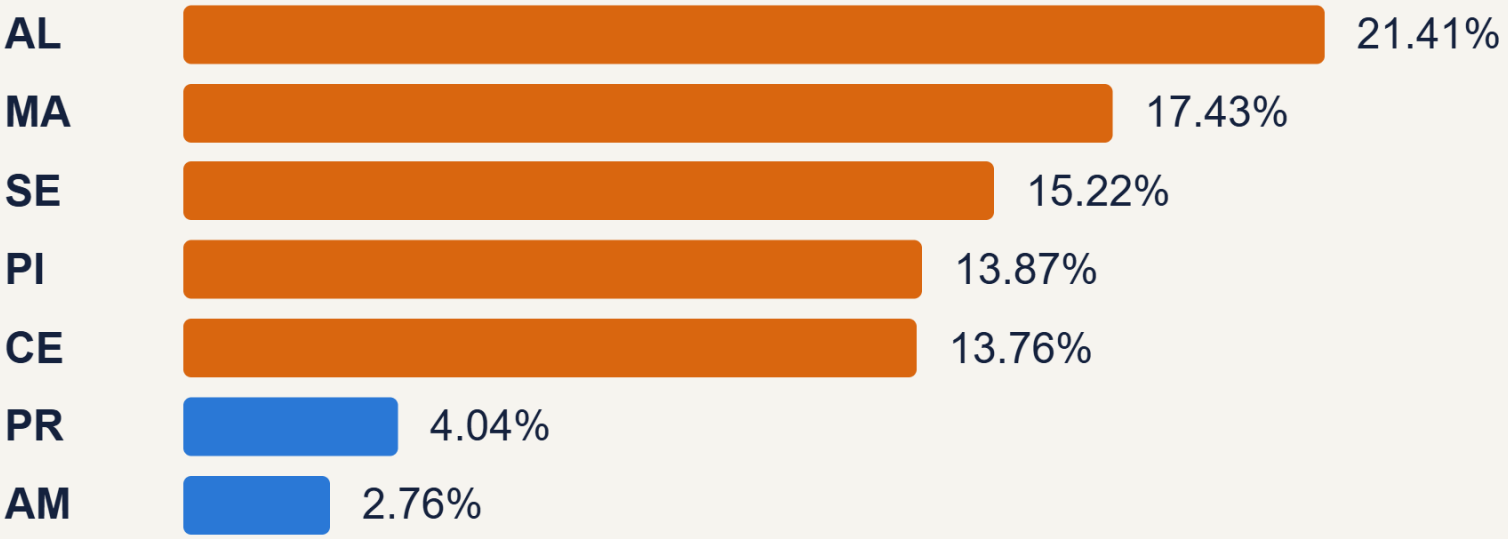
median time to spare against the promise

6.8%

arrived late: 2.9% Late (1-5 days), 3.9% Super Late (over 5)

Orders not delivered (2,971) are excluded. Days_Difference uses calendar dates.

Late deliveries are a Northeast problem



Nationwide late rate: 6.77%. Alagoas is about 3x the national rate.

Northeast states 335-1,279 orders each. Late % = Late + Super Late.

Distance from São Paulo does not explain it

0.06

Spearman correlation, distance vs late % (27 states)

2.8-3.8%

late in AC, AP, AM, RO, the farthest states (2,466-2,707 km)

13.8-21.4%

late in the five worst states, all closer (1,732-2,369 km)

Late % by region: Northeast 12.7, North 8.6, Central-West 6.5, Southeast 6.1, South 5.9

São Paulo as the hub is an assumption. Straight-line distance.

Late deliveries drive bad reviews

4.29

On Time: average review, 6.6% one-star

2.99

Late: average review, 33.6% one-star

1.74

Super Late: average review, 68.8% one-star

Caveat: roughly 37% of all one-star reviews come from late orders, so most come from on-time ones.

Reviewed orders only. Correlation of delay with score: 0.27.

Candidate's Choice: same buffer, different outcome

Far North: padded

AP promised 48 days, actual 25

AM promised 46, actual 26

AC promised 41, actual 18

RO promised 38, actual 18

20-23 days of padding, under 4% late

Northeast: buffer, still late

Median days early: AL 10, MA 11, SE 13, CE 13, PI 14

Paraná: 14 days early, 4.0% late

Late rates are 3-5x higher at a similar median buffer, which points to a slow tail rather than a low median.

Median promised vs actual days by state.

Recommendations

1. Focus carrier and last-mile fixes on AL, MA, SE, PI and CE.
2. Set estimates state by state, using the slow tail rather than the median.
3. Keep the far-North buffers under review: they are 20 days padded.
4. Look beyond delivery for one-star reviews: most come from on-time orders.

Caveat: small states such as RR (41 orders) are too small to rank.

Dashboard and notebook linked in the README.